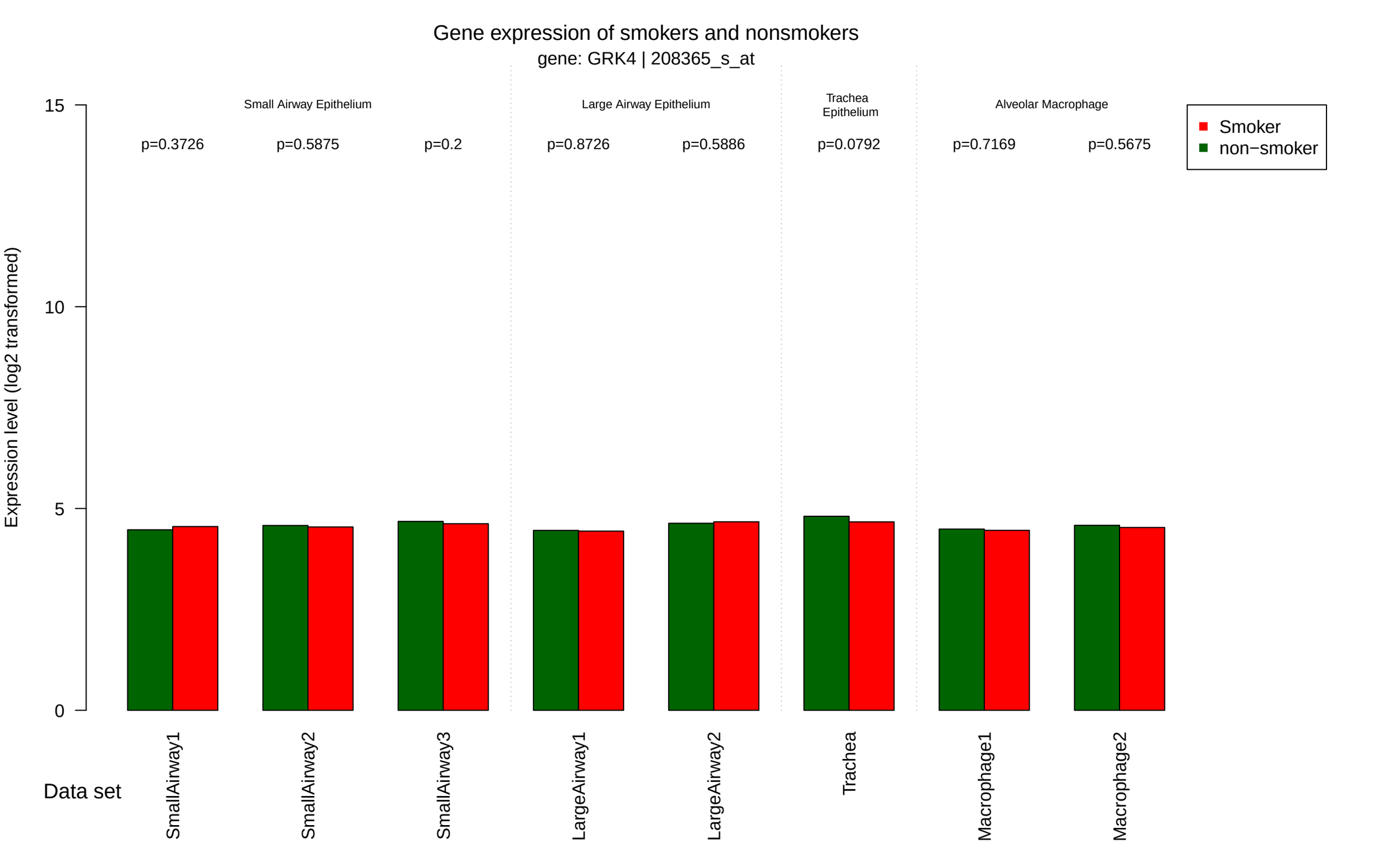
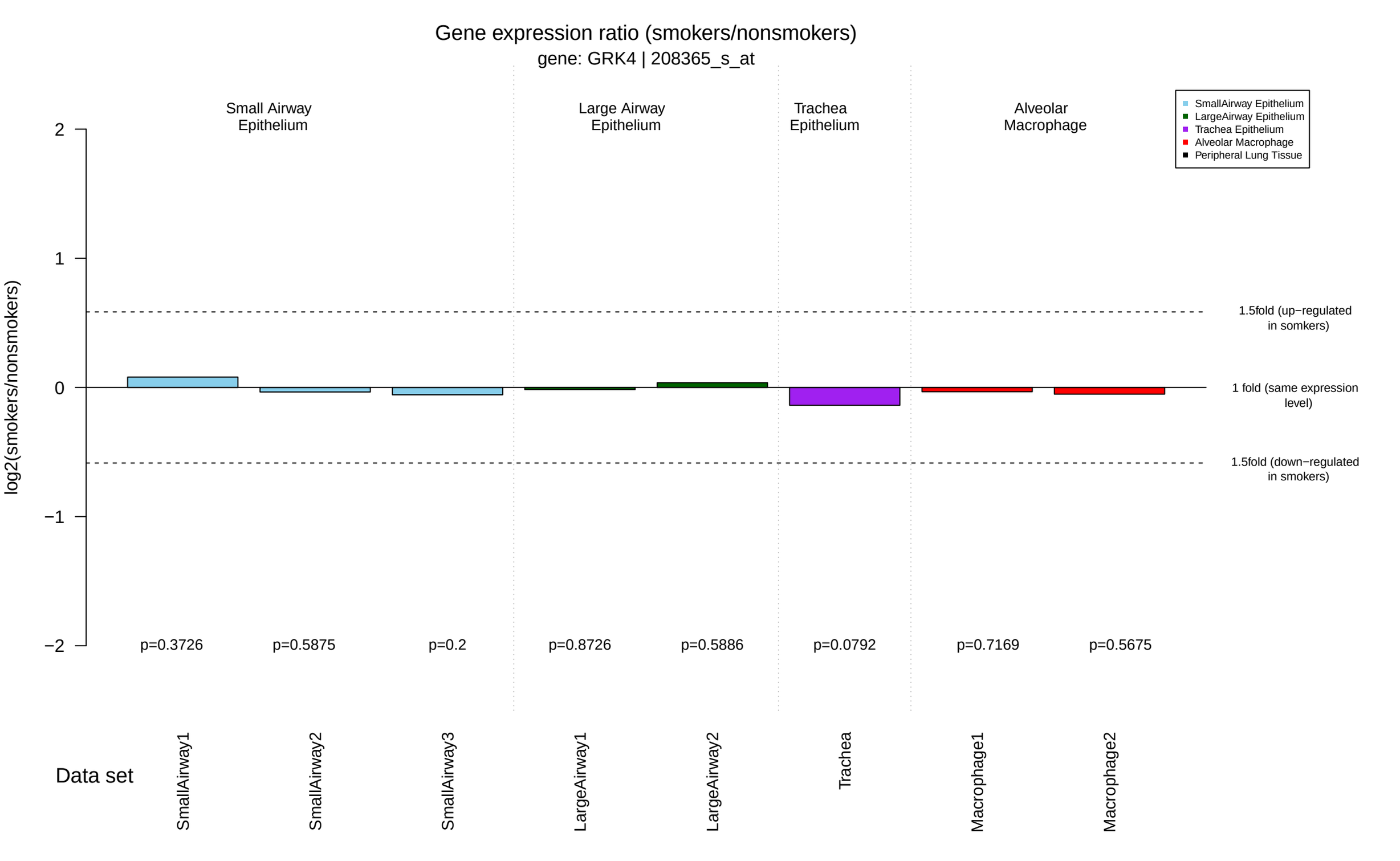
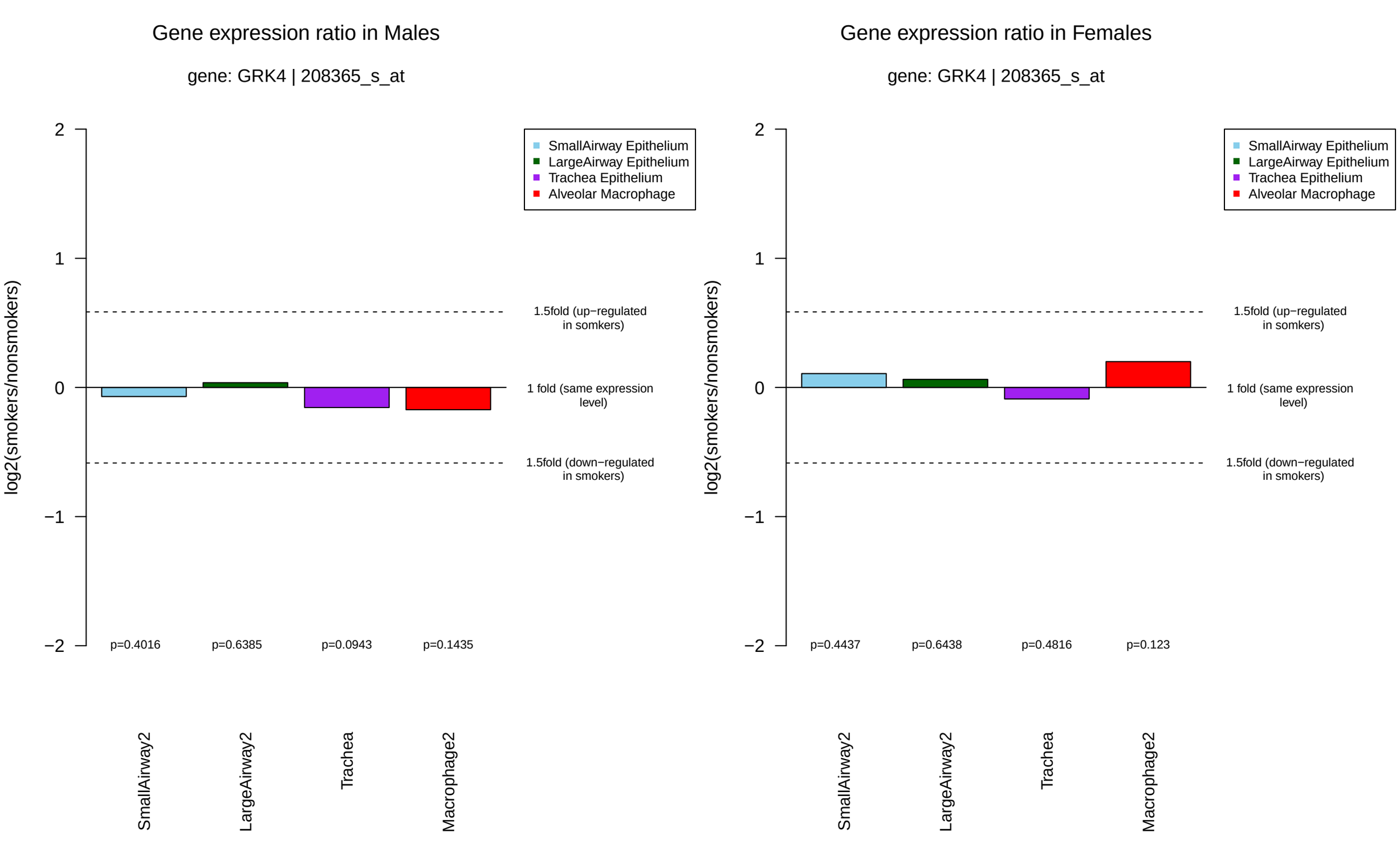
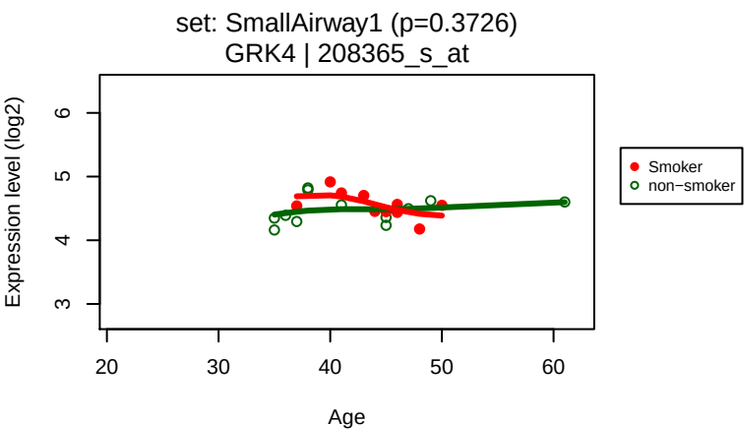
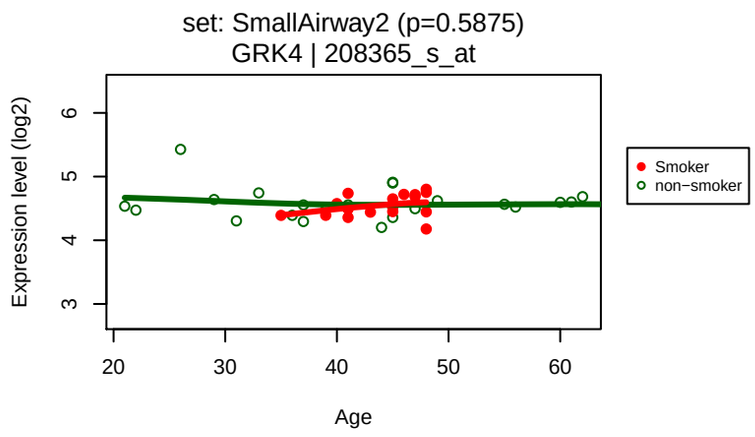
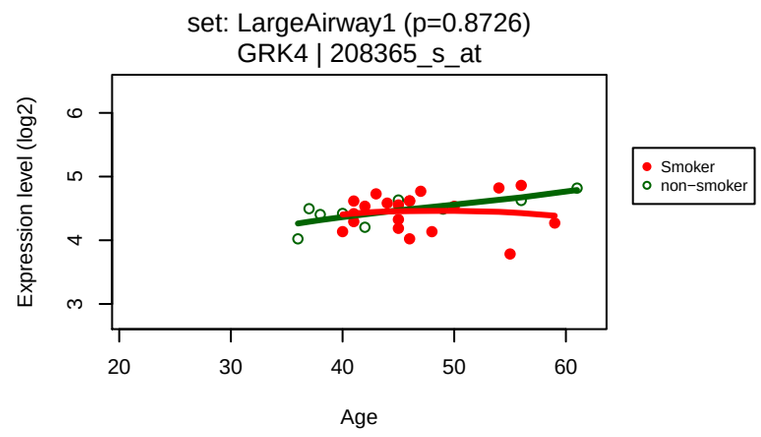

Gene expression of smokers and nonsmokers
gene: GRK4 | 208365_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3726
p=0.5875
p=0.2
p=0.8726
p=0.5886
p=0.0792
p=0.7169
p=0.5675
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

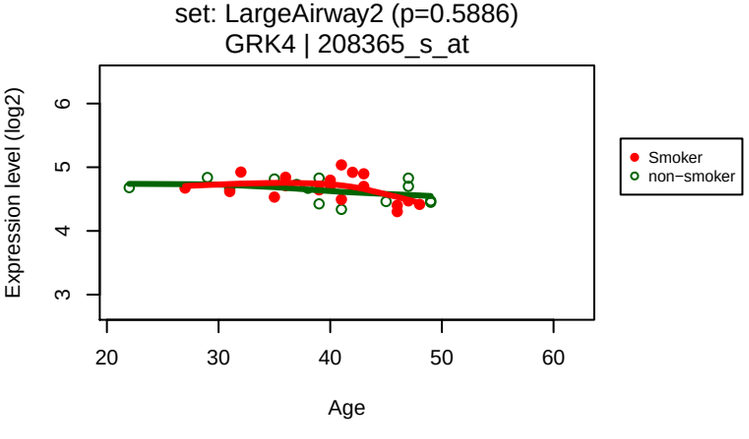

Gene expression ratio (smokers/nonsmokers)
gene: GRK4 | 208365_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3726
p=0.5875
p=0.2
p=0.8726
p=0.5886
p=0.0792
p=0.7169
p=0.5675
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

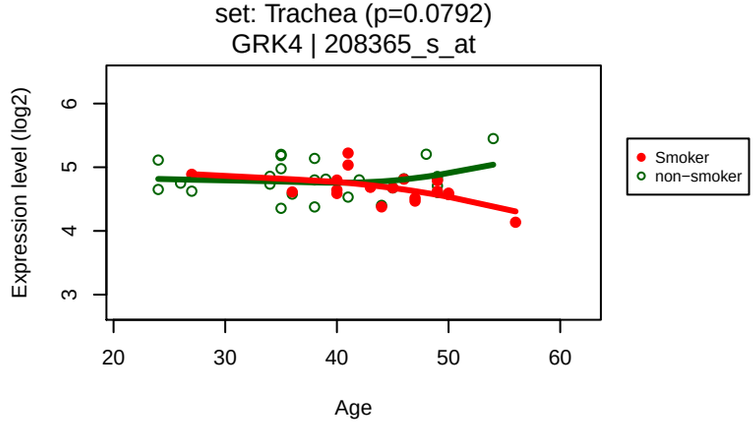

Gene expression ratio in Males
gene: GRK4 | 208365_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4016
p=0.6385
p=0.0943
p=0.1435
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRK4 | 208365_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4437
p=0.6438
p=0.4816
p=0.123
SmallAirway2
LargeAirway2
Trachea
Macrophage2

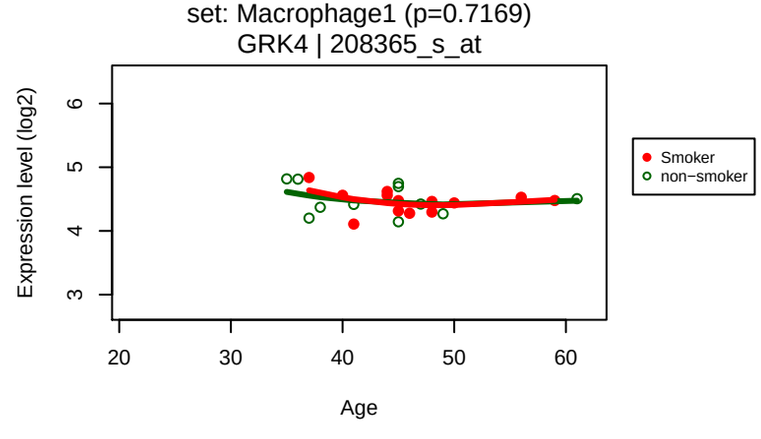

set: SmallAirway1 (p=0.3726)
GRK4 | 208365_s_at
Expression level (log2)
Age
Smoker
non−smoker

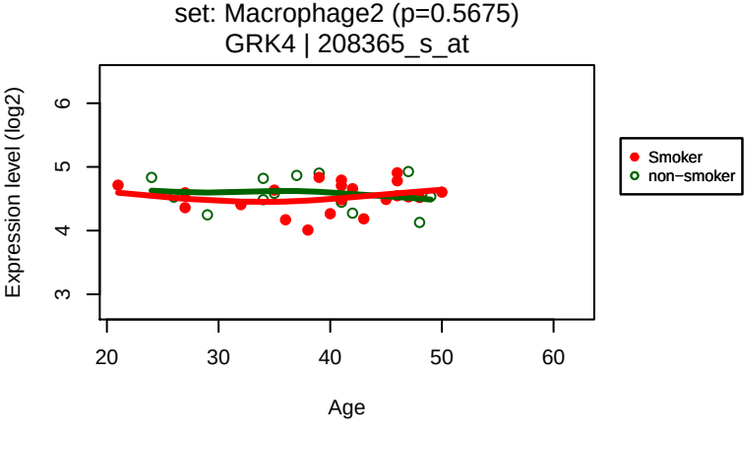

set: SmallAirway2 (p=0.5875)
GRK4 | 208365_s_at
Expression level (log2)
Age
Smoker
non−smoker

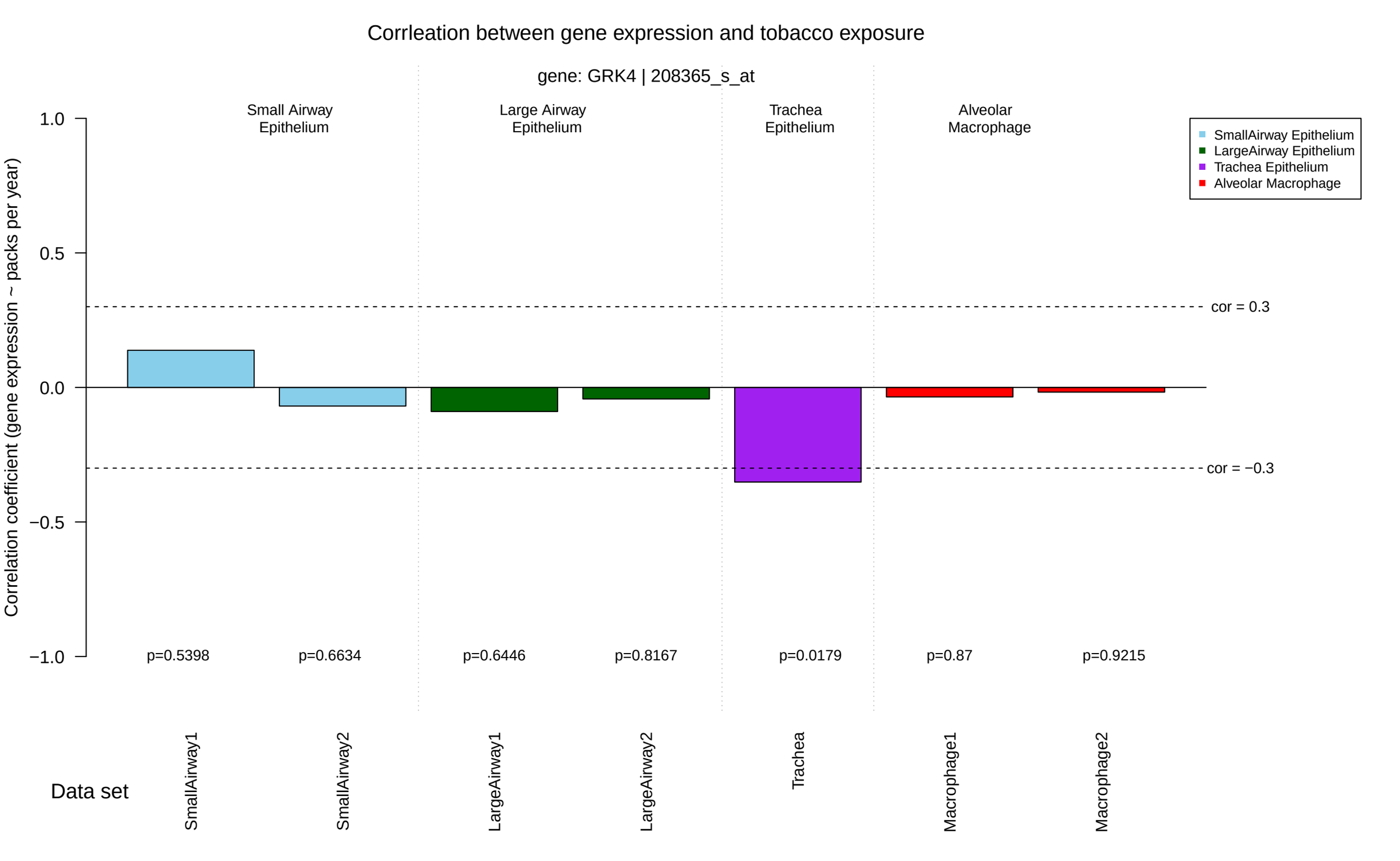

set: LargeAirway1 (p=0.8726)
GRK4 | 208365_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5886)
GRK4 | 208365_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0792)
GRK4 | 208365_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7169)
GRK4 | 208365_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5675)
GRK4 | 208365_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRK4 | 208365_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5398
p=0.6634
p=0.6446
p=0.8167
p=0.0179
p=0.87
p=0.9215
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2